

Conceptual Study of Real-time Ionospheric Threat Adaptation Using Space Weather Forecasting for GNSS Augmentation Systems

Jinsil Lee

*Korea Advanced Institute of Science and Technology**

Jiyun Lee*

Tetra Tech AMT

Sam Pullen

Stanford University

and

Seebany Datta-Barua

Illinois Institute of Technology

ABSTRACT

Current GNSS augmentation systems attempt to consider all possible ionospheric events in their computations of worst-case errors in user position solutions. The resulting error bounds are conservative because of this need to cover all possible anomalous conditions that might go undetected. This conservatism can be mitigated by subdividing anomalous conditions into several classes of severity and using different values of threat-model bounds for each class. This is possible if the level of ionospheric activity is classifiable and predictable (at least over periods of tens of minutes to hours) from measurable space weather conditions.

This paper presents a new concept of real-time ionospheric threat adaptation that adjusts the ionospheric threat model in real time instead of always using the same ‘worst-case’ threat model. This is done by utilizing a relationship between ionospheric activity and space weather indices. The worst-case threat is defined as a function of the values of space weather indices. Predicted values of space weather indices are used for determining the corresponding threat model. Since space weather prediction itself is not reliable due to prediction errors, an uncertainty model is derived from 15 years of historical data, and the prediction errors are bounded to the required level of integrity of the system being supported.

This concept was demonstrated by assessing the performance of real-time ionospheric threat adaptation when applied to the Local Area Augmentation System (LAAS) threat model used for Category I precision approach in the Conterminous United States (CONUS). The disturbance-storm time (Dst) index was selected as a measure of space weather intensity. The relationship between final Dst and the worst ionospheric gradients (or “slopes”) identified in CONUS was defined. The ‘predicted Dst bound’ derived by taking into account

prediction error statistics was then used to determine the worst slope bounds in real-time. As a result, the upper slope bound of the current threat model (425 mm/km for LAAS) could be lowered about 94% of the time during the 15 years of data (from 1995 to 2009) with an integrity of $(1 - 10^{-7})$ using the bounded prediction value of Dst for real-time threat determination.

1.0 INTRODUCTION

The ionosphere is one of the largest and most unpredictable error sources that can degrade the performance of Global Navigation Satellite Systems (GNSS). Many GNSS users therefore need to apply ionospheric corrections and integrity information using local or wide-area augmentation systems. The ionosphere, however, occasionally has strong variability that causes spatial and temporal ionospheric decorrelation between reference stations and users as well as small-scale ionospheric structures that generate scintillation. This variability becomes significant in the case of disturbed ionospheric conditions, resulting in decreased positioning accuracy and possible loss of integrity if undetected.

Ionospheric spatial gradients in the slant domain (i.e., along the actual path between satellite and receiver) of as large as 413 mm/km over baselines of 40 – 100 km have been observed in the United States during ionospheric storms since April 2000 [1]. Extreme gradients of this magnitude can pose a potential integrity threat to Local Area Augmentation System (LAAS) users because the residual range error at the CAT I decision height (DH) can be as large as 8 meters if undetected by the LAAS ground facility (LGF). It required the development of a worst-case ionospheric threat model for LAAS in the Conterminous U.S. (CONUS), the simulation of worst-case ionospheric errors for LAAS users, and the development of additional mitigation strategies [2], [3].

Figure 1 shows the current LAAS Category I (CAT I) ionospheric anomaly threat model for CONUS [1]. Anomalous spatial gradients are modeled as "fronts," or spatially linear semi-infinite wedges (parameterized by the slope of the ramp and its width) moving with constant speeds. The upper bound on the maximum gradient (or slope) of the front model as a function of satellite elevation angle was determined based on the worst gradients identified. This model was used for safety assessment and System Design Approval (SDA) of the Honeywell SLS-4000 LGF by the Federal Aviation Administration (FAA) for use in CONUS.

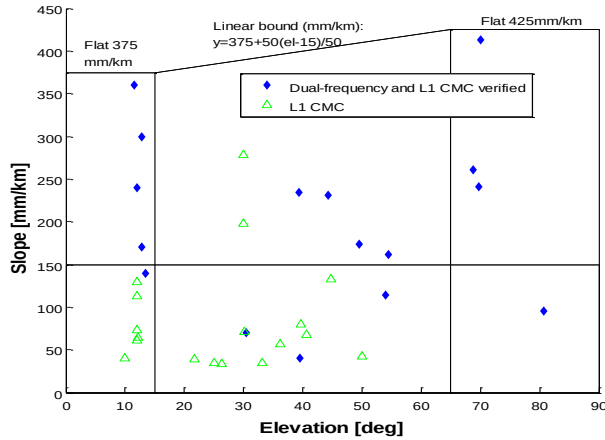


Figure 1 LAAS CAT-I ionospheric gradient threat space (Reproduction of Fig.13 of [1])

The upper bound of this threat model is used to simulate worst-case ionospheric errors for CAT I GBAS users by assuming the worst-case airborne satellite geometries and approach timing and direction relative to the LGF. These errors can be reduced to acceptable levels by inflating one or more integrity parameters broadcast by the LGF and consequently removing potentially hazardous satellite geometries. However, the CONUS threat model requires significant parameter inflation most of the time because it is driven by very rare events that cannot be guaranteed to be detected by the LGF. The largest observed gradients, which determined the flat slope bounds of 425 mm/km at high elevation and 375 mm/km at low elevation, were observed in Northern Ohio during the same ionospheric storm on 20 November, 2003. This was the most extreme event that occurred during the peak of the last solar cycle in CONUS. Threat observations populated in Figure 1 were observed and identified on only four days within 11-years period of the last solar cycle. Thus, always assuming the worst ionospheric condition within the threat space could lead to an excessive availability penalty. This conservatism can possibly be reduced if predictions of ionospheric activity indicating when ionospheric

conditions are less severe than the worst-case threat model bound are sufficiently reliable.

Many studies have shown a relationship between ionospheric activity and space weather indices. Mendillo, et al in [4] found that enhancements in total electron content (TEC) of the ionosphere coincided with increases in geomagnetic activity. Balan and Rao in [5] also showed a correlation between geomagnetic storm intensity and the strength of ionospheric storms by examining TEC and peak electron density data at both low- and mid-latitudes. However, the use of this information to potentially aid GNSS augmentation systems and users has not been thoroughly investigated. If the worst ionospheric threat can be related in advance to predicted space weather conditions, and the predictions are reliable to the required level of integrity, it would allow service to be enhanced.

This paper proposes utilizing this relationship in a new concept of real-time ionospheric threat adaptation. The ionospheric threat adaptation method selects threat model bounds in real time from a continuous function or from one of several pre-determined models, both of which are functions of predicted values of space weather indices. Section 2 introduces available space weather indices and selects Dst as a suitable index for real-time threat adaptation. The worst-case threat is then defined as a function of the selected space weather index. In Section 3, the uncertainty of space weather prediction is analyzed using historical data, and a method of bounding prediction errors is described. Methods of utilizing pre-analyzed information regarding bounded prediction errors to determine appropriate threat model bounds are proposed and expected performance is evaluated in Section 4. This study is concluded in Section 5.

2.0 RELATIONSHIP BETWEEN SPACE WEATHER AND IONOSPHERIC GRADIENT

2.1 Space weather indices

Real-time ionospheric threat adaptation utilizes external information on space weather conditions to determine the corresponding threat model bounds on a periodic basis (e.g., every few minutes). The first step required to realize this concept is to select a suitable space weather index (or indices) that is strongly correlated with the ionospheric threats of concern while also considering accessibility, fast update rate, short latency, long lead time, etc. Many organizations provide space weather indices in the form of 'real time' and/or 'prediction' values Table 1 lists three global space weather indices: disturbance, storm time

(Dst), planetary K (Kp), and interplanetary magnetic field (IMF) Bz, which are known to be strong correlated to ionospheric activity. The update rates of both real-time and prediction values and the lead time of prediction for indices provided by different organizations are presented in this table. "Lead time" indicates the length of time between the broadcast of predicted data and the period for which the prediction is intended to be valid. As an example, the University of Colorado provides predicted Dst updates every 10 minutes for the following hour at about 10 minutes past the hour. Therefore, the length of time from prediction generation to the future time that the predicted data references is about 50 minutes. In other words, the maximum "lead time" in which we can take advantage of the prediction is roughly 50 minutes.

Table 1 Global space weather indices

Parameters	Real time/ Prediction	Organization	Update rate	Lead time
Dst	Real time	WDC, Kyoto Univ. [6]	Hourly	
		USGS [7]	1 min / 1 hour	
	Prediction	Univ. Colorado [8]	10 minutes	50 minutes
		Univ. of Texas ¹⁾ [9]	10 minutes	2 hours
		NiCT, Japan [10]	Hourly	50 minutes
Kp	Real time	GFZ Potsdam [11]	3 hours	
		SWPC [12]	3 hours	
	Prediction	SWPC (WingKp)	15 minutes	1 hour / 4 hours
IMF Bz	Real time	SWPC	1 minute	

1) No digital data is available (only plots are provided).

In general, longer lead times and shorter update rates are desirable for the purpose of this study. Thus, predicted data from the University of Colorado at Boulder [8] and the University of Texas at Austin [9] are good candidates. However, no numerical data is publically available from [9]; thus the predicted Dst index from [8] was selected.

Real time values of the hourly Dst index are available at the World Data Center (WDC) for Geomagnetism at Kyoto University in Japan [6]. These 'real time' Dst values are typically revised after a few months and replaced with 'provisional' Dst. After about two years, the 'provisional' Dst is replaced with 'final' Dst. In this study, we assume that 'final Dst' represents the true value of Dst at the indicated times. Therefore, 'final Dst' values are used in Section 3 to derive an uncertainty model for the predicted values. In comparison with Kp, both final and predicted values of Dst are available for longer periods of time: from 1995 to 2009, inclusive, or a total of 15

consecutive years (this gives a total of 788,976 data points, as predicted values of Dst have a resolution of 10 minutes). The selected index, Dst, also has a strong correlation with ionospheric activity, as shown in the following subsection

2.2 Correlation analysis between space weather indices and ionospheric spatial gradients

As explained earlier, a strong relationship between space weather indices and ionospheric spatial gradient is needed to utilize adaptive threat models which vary depending on space weather intensity. Figure 2 shows day to day variations of Dst and the standard deviation of vertical (or zenith) ionospheric spatial gradients (σ_{vig}) from Day 206 to Day 214 in 2004. The " σ_{vig} overbound," which represents an inflated σ_{vig} to cover the non-Gaussian tails of the actual distribution (black circles) [13], was estimated on each day for the Continuous Operating Reference Stations (CORS) in CONUS separated by 80 to 100 km. The daily maximum 'final' Dst values are plotted at the center of each corresponding day (blue diamonds) along with hourly 'final' Dst values (the dashed line) provided by the WDC for Geomagnetism at Kyoto University. These variations show a similar trend, indicating that the quantities (Dst and σ_{vig} overbound) are highly related to each other. The correlation coefficient between " σ_{vig} overbound" and Dst was very high (greater than 0.8) during the period of interest, which includes both nominal and ionospherically active days, as shown in Figure 3 (see [14] for more details). These results support the use of Dst for real time threat adaptation.

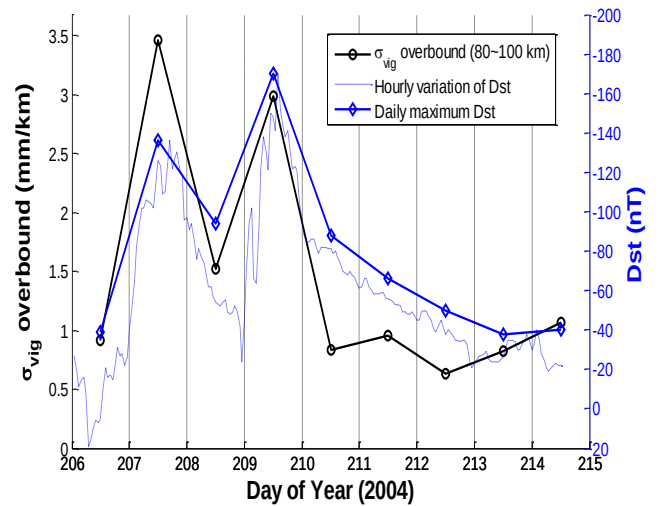


Figure 2 Variations of Dst index and standard deviation of vertical ionospheric gradients

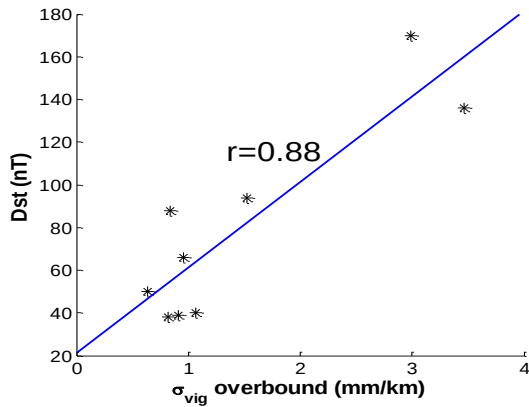


Figure 3 Correlation of Dst index and standard deviation of vertical ionospheric gradients

2.3 Worst case threat as a function of space weather index

This section defines the worst case threat (in terms of bounding ionospheric gradient) as a function of the values of the selected space weather index Dst. We first analyze the relationship between the ‘final’ Dst and ionospheric gradients. Anomalous ionospheric gradients observed in CONUS during the last solar maximum period (2000 – 2004) were used for this study. The ten dates from which data were collected and analyzed to generate the current CONUS threat model are listed in Table 2 along with Kp and Dst indices [1]. Eleven additional dates were selected by applying predetermined criteria (days where the daily maximum value of Kp is greater than six or that of Dst is less than -200nT [15][16]) to the data from 2000 to 2004, as shown in Table 3. In order to identify large ionospheric gradients, we used the Long Term Ionospheric Anomaly Monitoring (LTIAM) software package developed to build, validate and update threat models over the life cycle of the system by monitoring ionospheric behavior continuously [15]. This tool focuses on estimating ionospheric delays and ionospheric gradient magnitudes and automatically detecting gradients large enough to be hazardous to users [17][18].

Table 2 Dates investigated for the current CONUS threat model and space weather indices [1]

Day (UT mm/dd/yy)	Kp	Dst
04/06/00	8.3	-287
04/07/00	8.7	-288
07/15/00	9.0	-289
07/16/00	7.7	-301
09/07/02	7.3	-177
10/29/03	9.0	-350
10/30/03	9.0	-383
10/31/03	8.3	-307
11/20/03	8.7	-422
07/17/04	6.0	-76

Table 3 Additional dates selected and space weather indices

Day (UT mm/dd/yy)	Kp	Dst
08/12/00	7.7	-235
09/17/00	8.3	-201
03/31/01	8.7	-387
04/11/01	8.3	-271
04/12/01	7.3	-236
11/06/01	8.7	-292
11/24/01	8.3	-221
11/21/03	6.7	-309
11/08/04	8.7	-374
11/09/04	8.7	-214
11/10/04	8.7	-263

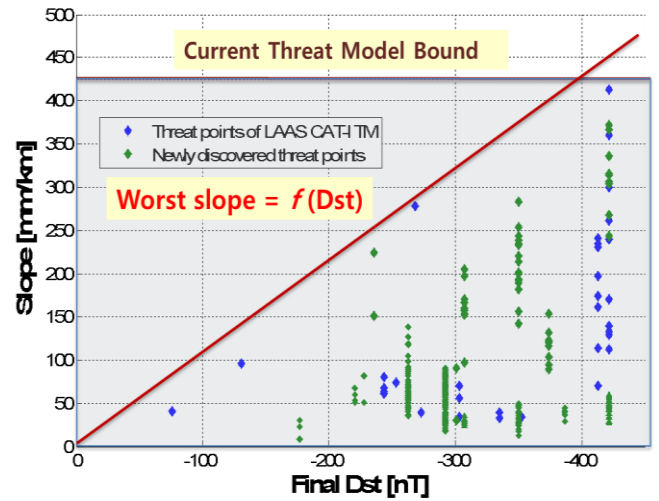


Figure 4 Ionospheric spatial gradient threat space: ionospheric gradient vs. ‘final’ Dst

Figure 4 shows the resulting threats (anomalous gradients) observed on the days listed in Tables 2 and 3. Observations of individual events populate the ionospheric spatial gradient threat space, with gradient or slope shown on the y-axis and the corresponding daily maximum value of ‘final’ Dst shown on the x-axis. The threat points identified previously (from days in Table 2) and included in the CONUS threat model are marked with blue diamonds (◆). The newly discovered threat points from this data analysis (from days both in Table 2 and Table 3) are denoted with green diamonds (◆). The newly discovered threat points include additionally observed threat points by LTIAM from the dates listed in Table 2 and threat points discovered from the dates listed in Table 3. Based on the identified threat observations, the maximum slope which bounds all ionospheric spatial gradients as a function of the ‘final Dst’ is determined and is shown as the red solid line.

3.0 UNCERTAINTY MODEL OF SPACE WEATHER PREDICTION

3.1 Space weather prediction errors

Space weather prediction is not free from errors which cause the predicted values to deviate from true values. Prediction errors exist due to several factors: 1) inaccurate and inappropriate input measurements (i.e., solar wind measurements) to a prediction model; 2) inaccuracies in a model due to a limitation on modeling magnetosphere; 3) the complex behavior of the space weather [19]. Figure 5 shows the comparison of ‘final’ Dst versus predicted Dst for the total number of 788,976 data points collected from 1995 to 2009. The predicted Dst values provided by the University of Colorado at Boulder are plotted on the horizontal (x) axis, and the ‘final’ Dst values provided by WDC at Kyoto University are plotted on the vertical (y) axis. Again, we assume that the ‘final’ Dst represents the true Dst. A linear relationship between the predicted Dst and the ‘final’ Dst is shown in the two-dimensional scatter plot. However, the predicted Dst does not always bound the ‘final’ Dst. In particular, approximately 32 percent of the predicted Dst values (those plotted in the lower, shaded triangle) are smaller than the corresponding ‘final’ Dst, meaning that the prediction was optimistic and could introduce safety concerns if used directly. Therefore, an uncertainty model for the predicted Dst values needs to be established.

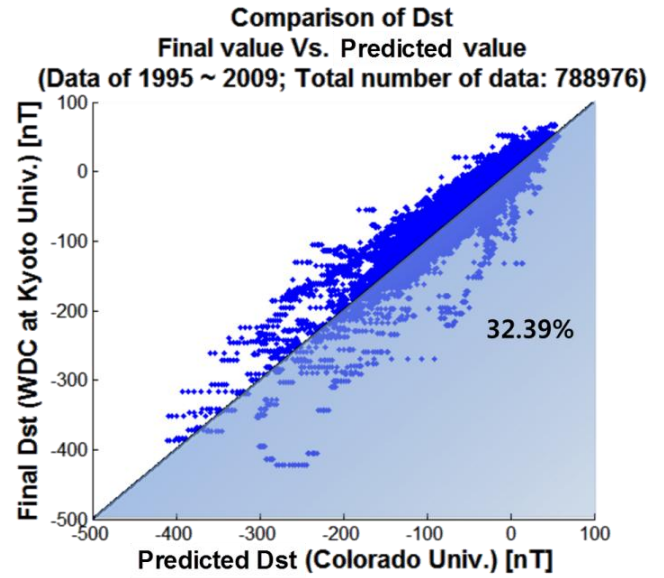


Figure 5 Statistics of space weather prediction uncertainty

To estimate the prediction uncertainty, we first determine the prediction errors, which are represented by ‘differential Dst (DDst)’ and are computed by subtracting the ‘final’ Dst from the predicted Dst at each epoch. Figure 6 shows the variations of computed DDst values over the fifteen consecutive years between 1995 and 2009. While most of the DDst values are close to zero, relatively large DDst values were observed from days on which ionospheric storms occurred (note that DDst values of 178.2nT and 178.9nT occurred on 6 November 2001 and 20 November 2003, respectively).

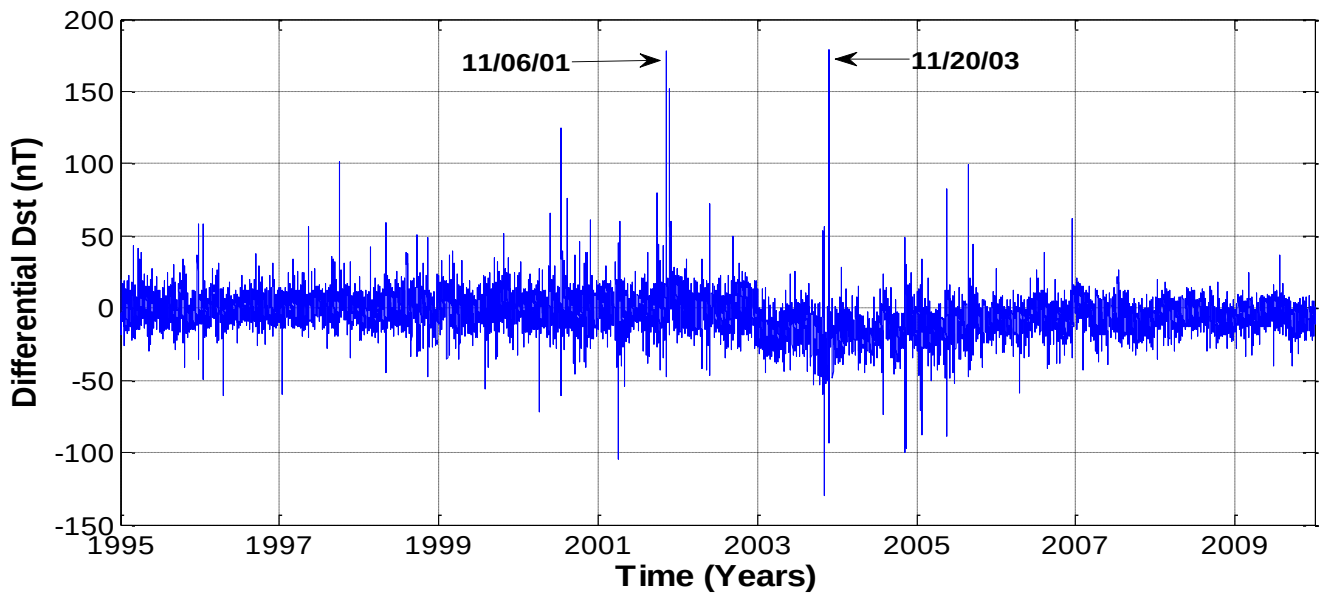


Figure 6 Differential Dst variations over time

3.2 Statistics of prediction errors

By using computed DDst values, a statistical analysis of prediction uncertainty is performed. First, DDst values are divided into 5 bins of predicted Dst in order to allow for variable behavior under different levels of ionospheric intensity while maintaining a sufficient number of samples per bin to obtain reliable statistics:

- bin 1: predicted Dst $\geq$ 0nT
- bin 2: 0nT $>$ predicted Dst $\geq$ -25nT
- bin 3: -25nT $>$ predicted Dst $\geq$ -75nT
- bin 4: -75nT $>$ predicted Dst $\geq$ -175nT
- bin 5: predicted Dst $<$ -175nT .

Second, the mean (μ_{DDst}) and the standard deviation (σ_{DDst}) of differential Dst (DDst) in each bin are computed and used to normalize the DDst. The estimated mean (the curve with triangles) and one sigma values (the curve with circles) of DDst are shown in Figure 7. Lastly, the “ σ_{DDst} overbound” (the curve with asterisks) is computed for each bin to take into account the non-Gaussian tails of the normalized DDst distribution. Detailed steps for σ_{DDst} overbounding method are described in the following paragraph.

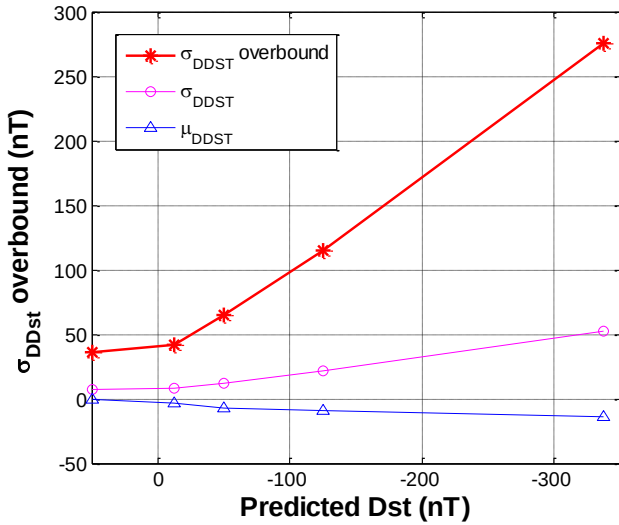


Figure 7 σ_{DDst} overbound result

Figure 8 shows the distribution of normalized DDst values on a logarithmic scale. DDst values are normalized by removing their means and dividing them by their standard deviations, which were computed separately for each bin. It is clearly seen that the distribution (the dotted curve) derived from the computed DDst shown in Figure 6 has non-Gaussian tails. It is noteworthy that the distribution has thicker and broader right side non-

Gaussian tails, where the predictions were optimistic, compared to the left side tails. If a zero-mean Gaussian distribution were assumed to model the prediction uncertainty, the nominal sigma (1σ) of a zero-mean Gaussian distribution (the dashed curve) would have to be inflated to cover the non-Gaussian tails of the actual distribution, especially non-Gaussian tails in the right side which may raise safety concerns. The inflation factor (f) bounding the distribution to the level of a probability of $1-10^{-5}$ is determined as shown in Figure 8 to take into account worst-case prediction errors. The ‘ σ_{DDst} overbound’ is then computed as $|\mu_{DDst}| + f\sigma_{DDst}$ for each bin by applying an inflation factor of approximately five as shown in Figure 7. It is seen that the determined ‘ σ_{DDst} overbound’ increases as the absolute value of the predicted Dst increases, implying that we need to consider larger uncertainty on prediction as the predicted Dst decreases.

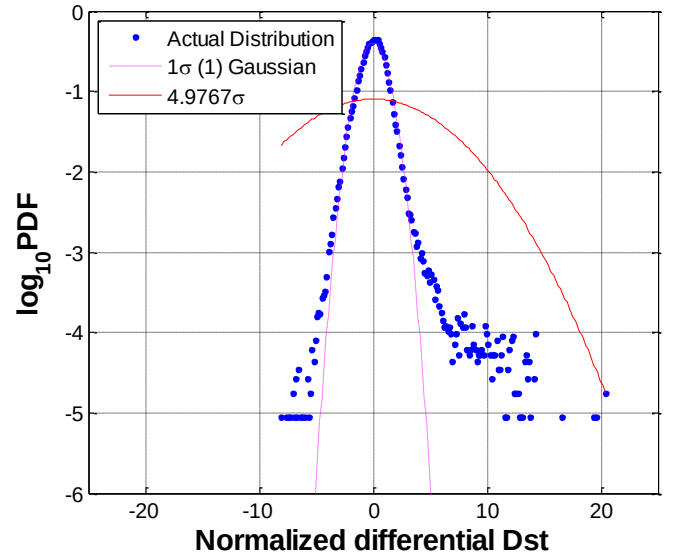


Figure 8 Distribution of normalized DDst

3.3 Bounding prediction errors to the required level of integrity

The resulting ‘ σ_{DDst} overbound’ from Section 3.2 is then used to compute bounded prediction errors to the required level of integrity by multiplying σ_{DDst} by a K factor. Here, the K factor is defined as the factor such that $P(-K \leq z \leq K) < x$, where z has a standard normal distribution and x denotes the required level of integrity. Figure 9 shows bounded prediction errors for each corresponding level of integrity from 0.68 ($K=1$, black curve) to $1-10^{-7}$ ($K=5.33$, red curve). In this study, we use a K factor of 5.33 to meet the required level of integrity of $1-10^{-7}$. The red line with asterisks shows the bounded prediction errors for the 10^{-7} integrity requirement as a function of predicted Dst.

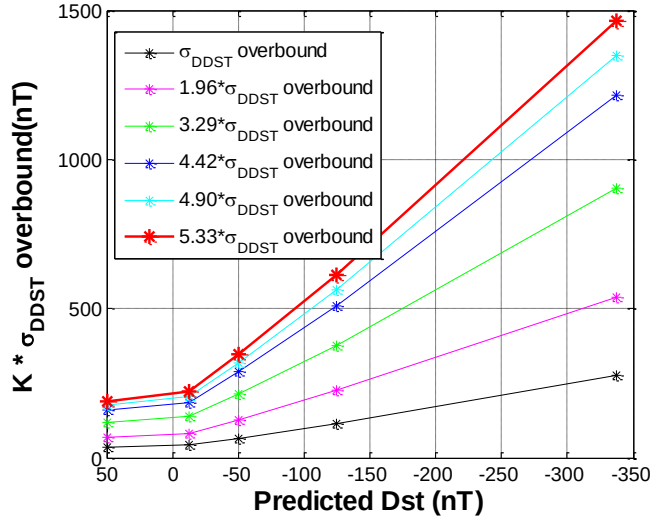


Figure 9 Bounded prediction errors

We then determine bounded predicted Dst, or the ‘predicted Dst_{bound}’, by subtracting the bounded prediction errors from the predicted Dst, as shown in Equation (1).

$$\text{Predicted Dst}_{\text{bound}} = \text{Predicted Dst} - K \cdot \sigma_{\text{DDst}} \text{ overbound} \quad (1)$$

Figure 10 shows the ‘predicted Dst_{bound}’ of given predicted Dst values for different levels of integrity. The bounded predicted Dst with a level of integrity of $1-10^{-7}$ (where K is equal to 5.33) is shown as the red curve with asterisks. This means that the ‘final Dst’ shall not exceed the ‘predicted Dst_{bound}’ with a probability of $1-10^{-7}$.

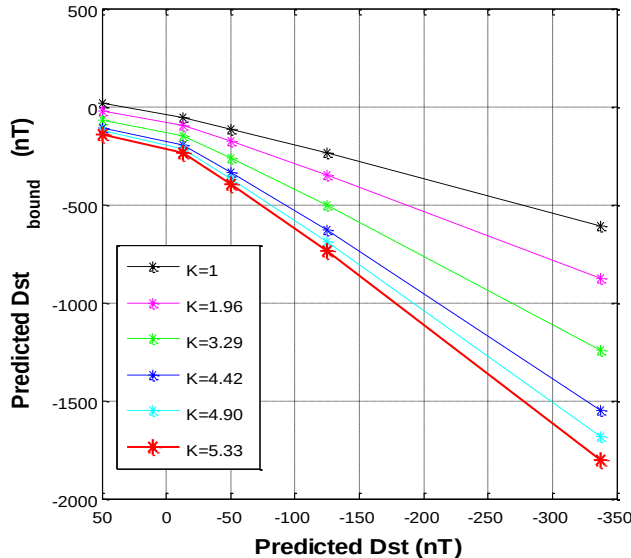


Figure 10 Predicted Dst_{bound} for different integrity levels

4.0 REAL-TIME IONOSPHERIC THREAT ADAPTATION

4.1 Threat model determination based on the bounded predicted Dst

Values of bounded predicted Dst with an integrity probability of 10^{-7} are used to determine corresponding threat bounds in real time based on pre-defined worst-case slope (or gradient) as a function of Final Dst. In Section 2.3, the worst slope was determined to have the value of 425mm/km when the Final Dst is -400nT or lower. Using our prediction uncertainty model, a predicted Dst of -50nT corresponds to a bounded predicted Dst of -400nT with 10^{-7} confidence. This suggests that the upper slope bound of 425mm/km in the LAAS CAT-I threat model for CONUS can be lowered if the predicted Dst is greater than -50nT (within the window of applicability of the prediction).

Figure 11 represents the slope (or gradient) bounds determined as a function of predicted Dst for several different integrity requirement (K-values) in consideration of the prediction uncertainty. The red dashed line shows the determined slope bound with an integrity probability of 10^{-7} ($K = 5.33$ from the standard Gaussian distribution). As expected, the current constant slope bound of 425mm/km can be lowered significantly when the predicted Dst value is greater than -50nT.

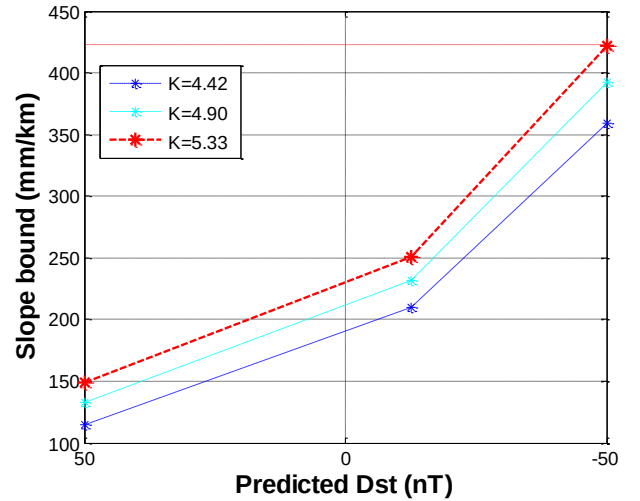


Figure 11 Slope (gradient) bound as a function of predicted Dst

4.2 Availability analysis of real-time threat adaptation

To see how much benefit can be obtained from the lowered slope bound using real-time threat adaptation, the distribution of predicted Dst over the 15-year data period was investigated. Figure 12 shows the histogram and

cumulative distribution function (cdf) of number of data points as a function of predicted Dst. The left-hand-side vertical axis represents the number of data points, while the right-hand-side vertical axis represents the corresponding cdf. One notable result from this figure is that approximately 94 percentages of predicted Dst values were greater than -50nT from 1995 through 2009, suggesting that real-time ionospheric threat adaptation would allow the worst-case slope bounds to be lowered about 94 percentage of the time.

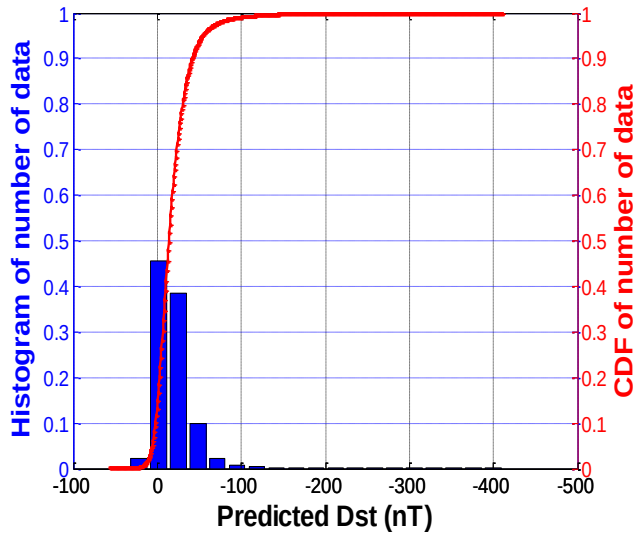


Figure 12 Histogram and cumulative distribution function of the number of data points as a function of predicted Dst

4.3 Improving prediction error bounding during low solar activity

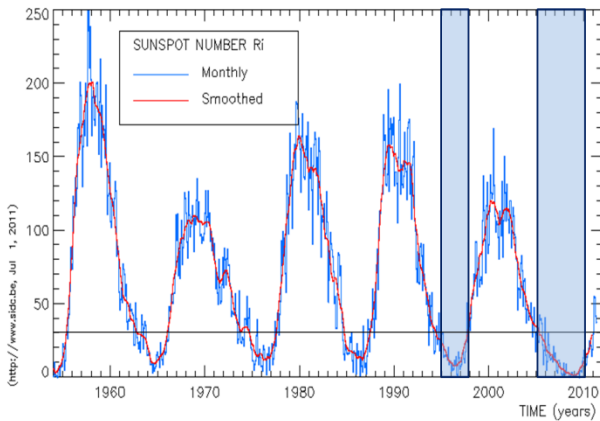


Figure 13 Solar cycle sunspot number progression [20]

This section investigates potential improvements in prediction error bounding by segregating data based on the overall level of solar activity. Figure 13 shows the solar cycle sunspot number progression from about 1950 to present, including the period analyzed [20]. Monthly

and smoothed sunspot numbers are represented over time. In this study, a maximum sunspot number of 30 is used as the criteria for distinguishing “solar minimum” periods of low solar activity (and low sunspot numbers). Among the total of 15 years in our database (from 1995 to 2009, inclusive), a total of 8 years from 1995 to 1997 and from 2005 to 2009 are inclusively selected as solar minimum years based on this criteria. These periods are highlighted with blue boxes in Figure 13.

Using only data during the 8 years of low solar activity, the same procedure was repeated in order to examine the difference in prediction error bounding. Figure 14 shows the distribution of normalized DDst derived by using only solar minimum period data. The non-Gaussian tails of normalized DDst are significantly thinner when only solar minimum period data are considered. This indicates that smaller inflation factors are needed for bounding the actual distribution. The resulting inflation factor needed for Gaussian bounding was reduced from 4.98 in Figure 8 (for all 15 years of data) to 3.98 in Figure 14.

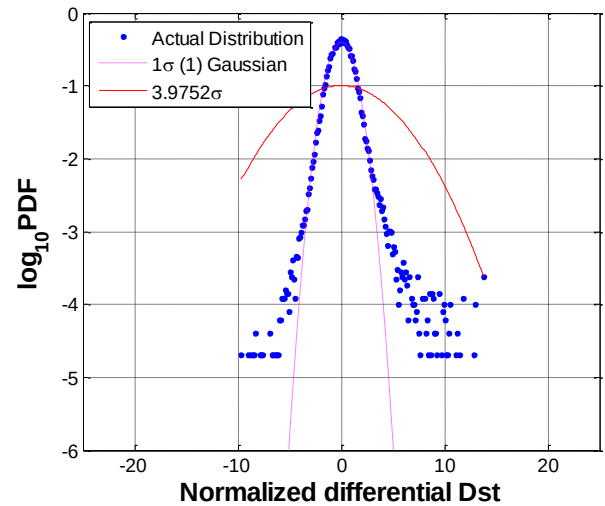


Figure 14 Distribution of normalized DDst using solar minimum period data

The corresponding bounded predicted Dst and slope bounds are compared in Figure 15 and Figure 16 with a blue dashed line (using all data) and a red dashed line (using only data during low solar activity). As the inflation factor is reduced by using only data during low solar activity, the bounded predicted Dst and slope bounds also decrease. This suggests that the conservatism of prediction error bounding over all solar conditions can be potentially reduced by segregating data by solar activity if this segregation can be done reliably and separately from space weather predictions.

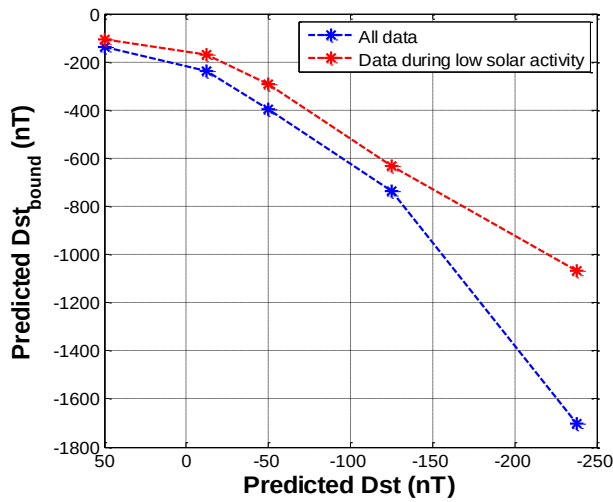


Figure 15 Comparison of predicted Dst_{bound}

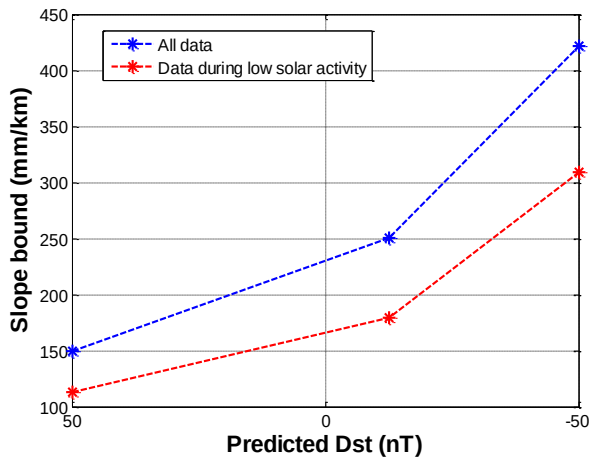


Figure 16 Comparison of slope bounds

4.4 Approaches for practical application of real-time ionospheric threat adaptation

The adaptive slope bounds shown in Figures 11 and 16 are represented only for the center points of each defined bin of predicted Dst used to derive these bounds. Therefore, great care must be taken in the practical application of real-time ionospheric threat adaptation. This paper suggests two potential approaches for applying these results in practice. First, linear interpolation between the determined slope bound points is the simplest way to formulate a continuous threat bound, and this is represented in Figure 17. A revised slope bounding any value of predicted Dst greater than or equal to -50nT can be determined by interpolation, as illustrated by the red dashed arrow in Figure 17. However, linear interpolation contains uncertainties because it cuts “in between” the center points that represent each bin of predicted Dst data. As a result, the predicted Dst bound derived for a single bin of data is not guaranteed to hold for a linear interpolation within the bin. . Therefore, the safety of this linear interpolation approach needs further validation.

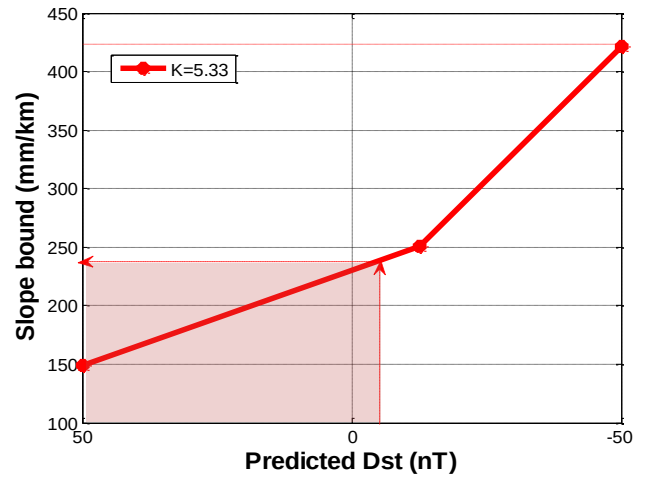


Figure 17 Linear interpolation method for practical operation of real-time threat adaptation

A second and more conservative method is to adopt the stepped threat slope bounds determined for each bin of predicted Dst. That is, we consider the determined slope bound of each bin as the slope bound for whole range of the corresponding bin of predicted Dst. For example, a slope bound of 250mm/km would apply for the whole range of the corresponding bin of predicted Dst from 0nT to -25nT , as shown in Figure 18. As a result, a total of three slope bounds of 150mm/km , 250mm/km and 425mm/km would be used depending on the predicted Dst value. This approach enables users to select a corresponding threat model in real time from one of several pre-determined models. However, this approach is more conservative than the linear interpolation approach, as the minimum value of predicted Dst for which the slope bound is reduced increases from -50nT to -25nT . This reduced range of predicted Dst will somewhat decrease the benefit of real-time threat adaptation from what was suggested in Section 4.2. Further consideration of the best methodology for applying real-time ionospheric threat adaptation is required.

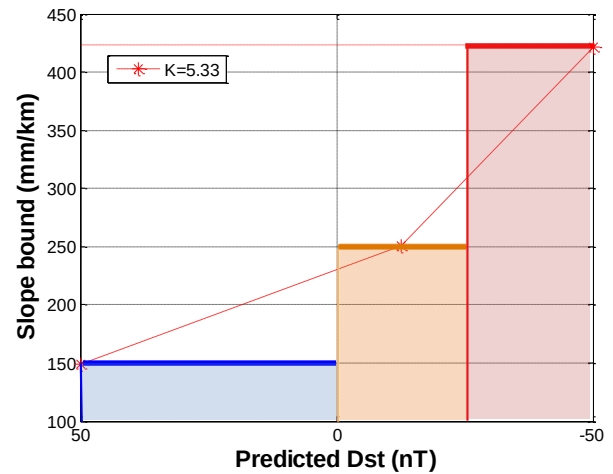


Figure 18 Stepped (bin-based) slope bound method for practical operation of real-time threat adaptation

5.0 CONCLUSION

Current GNSS augmentation systems contain a large amount of conservatism without making use of external information regarding ionospheric conditions. In this paper, a new concept of real-time threat adaptation using external space weather predictions is proposed to reduce conservatism. Using an appropriate predicted space weather index indicating ionospheric conditions sometime in the near future, it is possible to adjust the ionospheric threat model bounds in real time while assuring the required level of integrity. For this purpose, three steps have been performed in this study: 1) Finding suitable space weather metric; 2) analyzing the uncertainty of the space weather prediction; 3) determining adaptive threat model bounds based on bounded space weather prediction errors. These steps have been applied to the Ground-based Augmentation System (GBAS) and the LAAS CAT-I ionospheric threat model developed for (mid-latitude) CONUS locations. Performance assessment using the revised, adaptive LAAS CAT-I threat model shows that the upper bound of the current threat model (425mm/km) was lowered 94% of the time even though prediction error bounding with an integrity requirement of 10^{-7} considerably inflated the value of predicted Dst_{bound} and weakened the credit that can be taken for Dst predictions.

This real-time adaptation concept can improve the level of availability of precision approach operations for GBAS, and this possibility may extend to single-frequency-based Satellite-based Augmentation Systems (SBAS) as well. It is also applicable to the GBAS differentially corrected positioning service (DCPS), which provides differentially corrected latitude, longitude, velocity, and time for a range of applications beyond precision approach and landing. As space weather predictions become more available and reliable, their benefits to GNSS service performance should continue to grow.

For further study, more work is needed to develop and optimize a specific method for the operational implementation of real-time ionospheric threat adaptation that considers the inner workings of GBAS ground systems. Further investigation of the statistical bounding of space weather prediction errors is also advisable, as the difference between predicted and final Dst is clearly non-Gaussian in the tails. This can be done by using different probability models with wider tails, such as the Laplacian distribution, to bound the distribution of differential Dst . Lastly, more detailed evaluation of the performance and availability benefits from using adaptive ionospheric threat modeling is also planned.

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